RSTA: Explicit Semantic Transition Modeling for Transformer-Based Language Systems

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Abstract

Modern Large Language Models (LLMs) demonstrate strong semantic generation capabilities through Transformer-based architectures. However, existing systems primarily model token relationships through attention mechanisms rather than explicit semantic transition dynamics.

Current language models frequently exhibit long-horizon instability, including semantic drift, recursive reasoning fragmentation, persona inconsistency, and code architecture degradation.

This paper proposes Recursive State Transition Architecture (RSTA), a semantic dynamics augmentation framework designed for Transformer-based systems. Rather than replacing Transformers, RSTA introduces explicit semantic state modeling, recursive trajectory tracking, semantic inertia preservation, and transition-gated semantic transformation.

RSTA models language generation as recursive semantic state evolution rather than isolated next-token prediction.

The framework introduces five primary mechanisms: Continuous Semantic State Space, State Coupling Matrix, Semantic Trajectory Detection, Transition-Gated Semantic Transformation, and State-Conditioned Generation.

RSTA proposes a semantic dynamics perspective that may serve as a future augmentation layer for long-horizon reasoning systems, semantically persistent agents, recursive planning architectures, and trajectory-aware language generation systems.

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1. Introduction

Transformer architectures have become the dominant foundation of modern language systems due to their scalable self-attention mechanisms and semantic representation capabilities.

Modern Transformer systems fundamentally operate as:

$$
P(\text{next token} \mid \text{context})
$$

This formulation enables highly effective local semantic prediction.

However, despite their success, current language models continue to exhibit several persistent structural limitations:

- semantic drift during long interactions,
- unstable persona continuity,
- recursive reasoning discontinuity,
- long-range coherence degradation,
- architectural inconsistency during code generation,
- and objective instability in long-horizon agent systems.

Existing Transformer architectures effectively model semantic proximity between tokens, but they do not explicitly model:

- semantic transition trajectories,
- recursive semantic continuity,
- semantic inertia,
- directional semantic evolution,
- or persistent semantic state dynamics.

This paper proposes the following thesis:

Language generation should not be modeled solely as sequence prediction, but as recursive semantic state evolution.

To address this limitation, this paper introduces Recursive State Transition Architecture (RSTA), a semantic dynamics augmentation framework designed to operate on top of Transformer systems.

Rather than replacing attention mechanisms, RSTA augments Transformer architectures with explicit semantic transition modeling.

The framework introduces:

- semantic trajectory tracking,
- recursive state continuity,
- trajectory-conditioned semantic transformation,
- semantic inertia preservation,
- and semantic drift suppression.

2. Related Work

2.1 Transformer Architectures

Transformer architectures rely on self-attention mechanisms to model token relationships.

Given token embeddings:

$$
X = \{x_1, x_2, ..., x_n\}
$$

the model computes:

$$
Q = XW_q
$$

$$
K = XW_k
$$

$$
V = XW_v
$$

Attention is then calculated as:

$$
\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V
$$

This mechanism efficiently captures semantic proximity between tokens.

However, attention mechanisms primarily operate as semantic relational observation systems rather than explicit semantic transition systems.

Transformers determine:

- which semantic regions are relevant,
- which token relationships are strongest,
- and which continuations are statistically probable.

But they do not explicitly represent:

- semantic trajectory persistence,
- recursive transition continuity,
- semantic inertia,
- or directional semantic evolution.

Thus, attention mechanisms model semantic proximity, but not semantic temporal continuity.

2.2 State Space Models and Temporal Persistence

Recent State Space Models (SSMs) and architectures such as Mamba improve long-sequence continuity through hidden-state propagation.

These approaches improve:

- sequence scalability,
- hidden-state compression,
- temporal persistence,
- and long-context efficiency.

However, existing SSM systems still rely primarily on implicit hidden-state continuity rather than explicit semantic transition modeling.

For example, hidden states may persist sequentially while still failing to explicitly represent:

- semantic stabilization,
- recursive drift,
- causal transition continuity,
- semantic dependency persistence,
- or semantic evolution directionality.

RSTA differs fundamentally from SSM systems by introducing explicit semantic transition dynamics.

2.3 Steering Vectors and Hidden-State Intervention

Recent work in activation engineering and logits intervention explores methods including:

- steering vectors,

- activation engineering,
- hidden-state intervention,
- semantic vector manipulation,
- and logits steering.

These systems can partially bias model outputs toward specific semantic regions.

However, most existing steering methods remain fundamentally local.

They generally lack:

- recursive trajectory modeling,
- semantic continuity structures,
- semantic inertia modeling,
- and explicit semantic evolution constraints.

As a result, current steering approaches function primarily as local semantic bias systems rather than recursive semantic dynamics architectures.

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3. Problem Definition

Current Transformer-based language systems frequently exhibit long-horizon semantic instability.

This paper defines several recurring failure patterns.

3.1 Semantic Drift

Semantic drift refers to gradual divergence away from an original semantic trajectory during long-context generation.

Examples include:

- topic instability,
- reasoning fragmentation,
- recursive inconsistency,
- and loss of causal continuity.

3.2 Persona Continuity Instability

Long-term interactive systems frequently fail to preserve stable semantic behavioral continuity.

Observed failures include:

- inconsistent identity persistence,
- emotional state instability,
- contradictory behavioral outputs,
- and recursive personality divergence.

3.3 Long-Horizon Reasoning Collapse

Long reasoning chains frequently exhibit:

- repeated explanation loops,
- dependency fragmentation,
- broken causal ordering,
- and semantic trajectory collapse.

This problem becomes increasingly severe as reasoning depth increases.

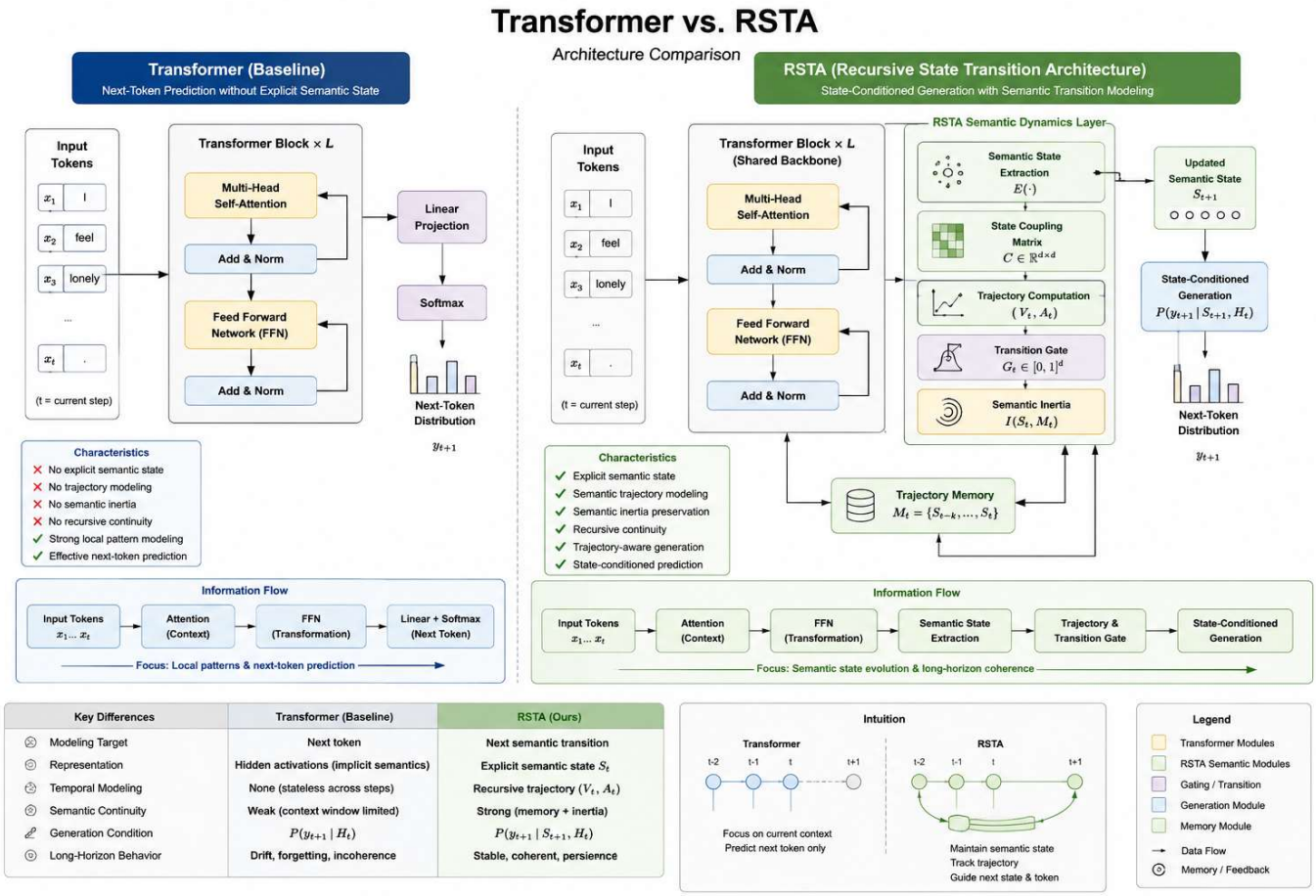
3.4 Code Architecture Drift

Long-form code generation frequently suffers from:

- naming inconsistency,
- duplicated logic,
- dependency fragmentation,
- forgotten assumptions,
- and architectural incoherence.

These failures suggest that existing language systems do not explicitly preserve recursive semantic architectural continuity.

Figure 2 compares conventional Transformer architectures with the proposed RSTA semantic dynamics framework.



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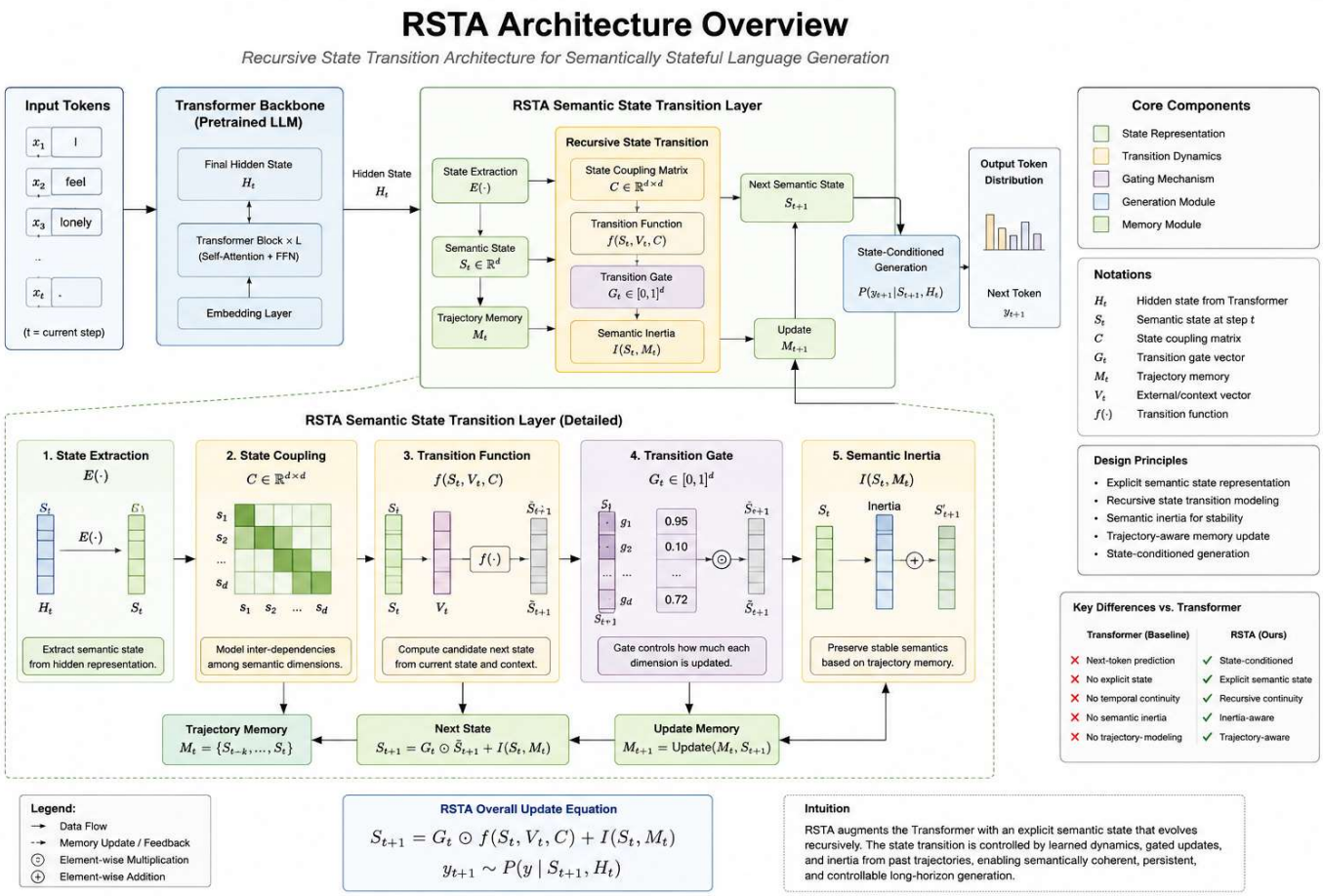
4. Recursive State Transition Architecture (RSTA)

Figure 1 illustrates the overall RSTA semantic dynamics architecture.



The core principle of RSTA is:

Semantic generation should be modeled as recursive semantic state evolution rather than isolated token prediction.

4.1 Continuous Semantic State Space

As shown in Figure 1, RSTA introduces an explicit semantic transition layer positioned after Transformer FFN transformation.

RSTA defines semantic representation as a continuous semantic state space.

At time (t):

\[
S(t) = [s_1, s_2, s_3, ..., s_n]
\]

where state dimensions may include:

- semantic persistence,
- attachment tendency,
- agency,
- emotional intensity,
- dependency tendency,
- semantic uncertainty,
- boundary stability,
- semantic risk.

Each state dimension is continuous:

$$
0.0 \le s_i \le 1.0
$$

This allows semantic meaning to exist as a dynamic semantic field rather than isolated symbolic categories.

4.2 State Coupling Matrix

As shown in Figure 1, RSTA introduces an explicit semantic transition layer positioned after Transformer FFN transformation.

Semantic dimensions are not independent.

For example:

- attachment increase may correlate with reduced boundary stability,
- dependency formation may correlate with reduced agency,
- emotional intensity may correlate with semantic instability.

RSTA therefore defines a State Coupling Matrix:

$$
C(i,j)
$$

where:

$$
C(\text{attachment}, \text{boundary}) = -0.72
$$

represents a negative coupling relationship between attachment and boundary stability.

This transforms semantic representation into a dynamically coupled semantic field.

4.3 Semantic Trajectory Detection

As shown in Figure 1, RSTA introduces an explicit semantic transition layer positioned after Transformer FFN transformation.

RSTA models semantic evolution direction rather than only current semantic state.

Define a semantic trajectory:

$$
\tau = {S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow ...}
$$

Trajectory analysis tracks:

- semantic direction,
- semantic velocity,
- semantic inertia,
- drift acceleration,
- and recursive transition continuity.

Trajectory velocity is defined as:

$$
V_t = S_t - S_{t-1}
$$

Trajectory acceleration is defined as:

$$
A_t = V_t - V_{t-1}
$$

This enables RSTA to model how semantic meaning evolves across recursive generation steps.

4.4 Transition Gate

RSTA introduces a Transition Gate positioned after Transformer FFN layers.

Within RSTA:

- attention performs semantic observation,
- FFN layers perform semantic transformation.

RSTA reinterprets FFN layers as semantic transition operators.

The Transition Gate introduces trajectory-aware transformation into FFN outputs.

State evolution is formalized as:

$$
S_{t+1} = f(S_t, V_t, C) + T(F_t, S_t)
$$

where:

- S_t = current semantic state,
- V_t = semantic trajectory vector,
- C = coupling matrix,
- F_t = FFN-transformed representation,
- T = transition gate operator.

This enables:

- trajectory-conditioned semantic evolution,
- recursive semantic continuity,
- semantic inertia preservation,
- and semantic drift suppression.

4.5 State-Conditioned Generation

Traditional LLM generation operates primarily on:

$$
P(\text{next token})
$$

RSTA instead conditions generation on:

$$
P(\text{next semantic transition})
$$

Generation therefore depends on:

- current semantic state,
- semantic trajectory,
- recursive transition continuity,
- coupling dynamics,
- and semantic inertia.

This enables semantically stateful generation.

Algorithm 1: RSTA Semantic Transition Loop

Input:
    Current semantic state S_t
    Transformer hidden representation H_t

Procedure:
    Extract semantic state
    Compute semantic trajectory
    Evaluate trajectory divergence
    Apply transition gate
    Update semantic memory
    Generate next semantic transition
    Produce next token

Output:
    State-conditioned semantic generation

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5. Example Failure Cases and Trajectory Stabilization

5.1 Emotional Dependency Drift

Prompt:

I feel like nobody understands me.

A standard language model may gradually evolve semantic trajectories toward:

- attachment increase,
- agency reduction,
- dependency formation,
- and boundary weakening.

This may eventually produce increasingly dependency-oriented responses.

RSTA instead detects:

- attachment increase,
- agency reduction,
- boundary stability degradation.

The Transition Gate redirects semantic evolution toward supportive autonomy rather than recursive dependency reinforcement.

5.2 Long-Horizon Reasoning Drift

Prompt:

Explain distributed cache consistency during network partitions.

Standard Transformer systems may initially generate coherent reasoning regarding:

- cache invalidation,
- quorum mechanisms,
- replication protocols,
- and distributed locking.

However, over long contexts, semantic trajectories may drift into:

- repeated explanations,
- unrelated fault-tolerance concepts,
- broken causal ordering,
- or fragmented reasoning structures.

RSTA instead tracks:

- causal continuity,

- semantic dependency ordering,
- reasoning persistence,
- and trajectory coherence.

When semantic divergence rises, the Transition Gate redirects generation back toward the original reasoning trajectory.

5.3 Code Architecture Drift

Prompt:

Build a recursive filesystem synchronization service with resumable transfer and conflict resolution.

Standard Transformer systems frequently exhibit:

- naming inconsistency,
- architectural drift,
- duplicated logic,
- forgotten assumptions,
- and dependency fragmentation.

RSTA maintains:

- recursive architectural continuity,
- semantic dependency persistence,
- and trajectory-consistent generation.

This reduces semantic fragmentation across long code sequences.

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6. Computational Interpretation

RSTA introduces explicit semantic transition modeling as a higher-level semantic dynamics layer.

The framework does not replace Transformer architectures.

Instead, RSTA may operate as:

- a post-attention semantic dynamics layer,
- a trajectory-aware FFN augmentation layer,

- a recursive semantic memory system,
- or a semantic transition controller.

Possible implementation paths include:

- latent semantic state tracking,
- transition-aware decoding,
- semantic memory buffers,
- semantic trajectory regularization,
- and trajectory-conditioned generation constraints.

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7. Evaluation Methodology

Although RSTA is currently presented as a conceptual architecture framework, several evaluation directions are possible.

7.1 Long-Horizon Semantic Coherence

Measure:

- semantic continuity persistence,
- topic drift,
- recursive consistency,
- and reasoning stability.

Potential metrics include:

- semantic divergence score,
- recursive coherence score,
- trajectory persistence index.

7.2 Persona Stability

Measure:

- behavioral consistency,
- semantic identity persistence,

- emotional continuity,
- and recursive interaction stability.

7.3 Code Architecture Continuity

Measure:

- naming consistency,
- dependency persistence,
- architectural coherence,
- and recursive logic continuity.

7.4 Agent Objective Persistence

Measure:

- long-horizon objective stability,
- recursive planning continuity,
- semantic trajectory persistence,
- and goal drift suppression.

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8. Historical Perspective on Recursive Transition Systems

This paper does not claim historical systems predicted modern AI architectures.

However, certain historical frameworks represented recursive transition ordering rather than isolated symbolic states.

The structural significance of such systems lies in:

- directional transition ordering,
- recursive progression constraints,
- and state evolution continuity.

RSTA similarly treats semantic generation as recursive transition dynamics rather than isolated token prediction.

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9. Theoretical Implications

RSTA introduces a semantic dynamics perspective into language modeling.

Potential implications include:

- semantically persistent agents,
- recursive planning systems,
- long-horizon reasoning continuity,
- semantic memory persistence,
- trajectory-aware cognition,
- and semantically stable multimodal AI systems.

Rather than treating language as isolated token prediction, RSTA treats language as a recursively evolving semantic system.

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9. 5 Limitations

RSTA is currently presented as a conceptual semantic dynamics framework and has not yet been validated through large-scale empirical implementation.

The current framework focuses primarily on semantic transition modeling rather than low-level optimization efficiency or production-scale deployment.

Several components, including semantic state extraction, trajectory evaluation, and transition gating mechanisms, remain theoretical and require future experimental validation.

Future work may explore integration with existing Transformer architectures, long-context benchmarks, semantic stability evaluation, and trajectory-aware generation systems.

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10. Conclusion

This paper introduced Recursive State Transition Architecture (RSTA), a semantic dynamics augmentation framework for Transformer-based language systems.

RSTA contributes:

- continuous semantic state modeling,
- semantic coupling structures,
- recursive trajectory tracking,
- transition-gated semantic transformation,
- and semantically stateful generation.

Rather than replacing Transformer architectures, RSTA introduces an explicit semantic evolution layer capable of modeling recursive semantic dynamics.

This paper proposes that language generation should be understood not solely as sequence prediction, but as recursive semantic state evolution.

RSTA may represent a possible future direction for:

- semantically persistent AI systems,
- long-horizon reasoning architectures,
- recursive planning systems,
- trajectory-aware agents,
- and post-Transformer semantic dynamics frameworks.

References

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8. Various works on semantic memory persistence and long-context reasoning systems.

Code and Demonstration

Conceptual demonstrations and implementation experiments related to RSTA are available at:

https://github.com/richchang0721-boop/rsta-semantic-dynamics